

Bijan Pourriahi

Senior Backend / Platform Engineer

Rails · PostgreSQL · Python · Rust · AWS · Linux · AI Coding Agents

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PROFILE

Senior backend/platform engineer with 10+ years of experience building SaaS products, financial systems, research platforms, internal tools, APIs, dashboards, data pipelines, and production infrastructure. Strong in Rails, PostgreSQL, Python, Rust, deployment automation, monitoring, AI-assisted development workflows, and data-heavy operational systems.

Most effective in small teams and high-ambiguity environments where complex workflows need to become reliable, maintainable production software.

SELECTED IMPACT

- Built proprietary financial systems infrastructure for research, simulation, account monitoring, broker integrations, risk controls, dashboards, data pipelines, deployment automation, and production observability.
- Designed high-speed simulation workflows capable of evaluating 1,000+ strategy/configuration variants and running year-long 1-minute-bar backtests in roughly 1-2 seconds.
- Led post-acquisition integration of five acquired SaaS systems at FocusVision, coordinating engineers and sysadmin teams across data migrations, infrastructure changes, production releases, and customer-impacting fixes.
- Owned end-to-end delivery of production software in small teams, including architecture, APIs, Rails applications, PostgreSQL data models, deployments, monitoring, incident response, and internal tools.
- Improved Solr/search indexing performance by roughly 100x by replacing a slow indexing workflow with a custom Java pipeline.

CORE SKILLS

Backend/platform: Ruby on Rails, API design, PostgreSQL, background jobs, service architecture, production debugging.

Infrastructure: Docker, AWS, Linux, CI/CD, deployments, monitoring, Grafana, Prometheus.

Data systems: PostgreSQL, ClickHouse, ETL, analytics pipelines, simulation workflows, performance tuning.

Product delivery: SaaS features, internal tools, dashboards, admin systems, workflow automation, technical leadership.

AI-assisted development: 2+ years using Codex, Claude Code-style coding agents, and AI-assisted workflows for implementation planning, code generation, refactoring, debugging, test creation, documentation, and repository-scale maintenance.

PROFESSIONAL EXPERIENCE

Private Trading Team / QuantBox — Contract Senior Backend / Platform Engineer

2021-Present | Remote

- **Context:** Contracted to build proprietary financial systems software for research, simulation, broker integrations, account monitoring, risk controls, dashboards, deployment workflows, and production operations.
- Designed and built a DB-backed control plane for managing account state, execution workflows, broker integrations, risk checks, monitoring, and operator actions.
- Built high-speed research and simulation workflows capable of evaluating 1,000+ strategy/configuration variants and running year-long 1-minute-bar backtests in roughly 1-2 seconds.
- Built dashboards, replay tools, and evidence workflows for reviewing system behavior, comparing configurations, and improving operational decisions.
- Built production/staging infrastructure with release bundles, health checks, alerts, Grafana/Prometheus monitoring, and Postgres-to-ClickHouse replication.
- Used Codex, Claude Code-style coding agents, and AI-assisted workflows for research planning, implementation scaffolding, refactoring, debugging, documentation, test generation, and large-repo maintenance while retaining code review and architecture ownership.
- Worked across Rust, Python, PostgreSQL, ClickHouse, Docker, Linux, WebSockets, data pipelines, and production infrastructure.

Rare Isle Fund — Principal Software Engineer

Apr 2019-Jun 2021 | Remote

- **Context:** Small hedge fund; principal engineer responsible for core automation, analytics, dashboards, and production operations.
- Built automated trading, research, analytics, and operations infrastructure from scratch, replacing manual research and strategy review with repeatable software workflows.
- Designed large-scale backtesting and strategy evaluation pipelines capable of processing millions of strategy variants daily, helping the fund test and compare ideas faster.
- Built live trading systems, automated selection workflows, analytics dashboards, operational controls, and monitoring tools that improved visibility into deployed systems, fund resources, and executed trades.
- Owned implementation and operations across Python, Pandas, Go, AWS, Grafana, Linux, DevOps, data pipelines, and production infrastructure.

FocusVision — Senior Software Engineer

Nov 2015-Jun 2018 | Portland, OR

- **Context:** Market research SaaS company; acquired product environment with multiple systems and integration needs.
- Led engineering work to integrate five acquired companies/systems into FocusVision, consolidating multiple Rails/PostgreSQL systems into a more unified production platform for customer and internal workflows.
- Ran and mentored engineering teams while delivering SaaS features, integration work, bug fixes, and production improvements.
- Managed deployments and coordinated with sysadmin/operations teams across production releases, infrastructure changes, and integration work.
- Built and maintained customer-facing features, admin tools, APIs, database-backed workflows, internal operations systems, deployment automation, and release processes.
- Owned architecture and delivery work across data migration, production support, release workflows, and cross-team technical coordination.

ADDITIONAL EXPERIENCE

24tru — Software Engineer

Apr 2013-Jun 2015 | Lakewood, CO

- **Context:** Small market research SaaS team building software for research collection and video-based workflows.
- Co-owned the company's software platform with one other developer, covering production maintenance, data migrations, new features, customer-impacting fixes, and day-to-day technical operations.
- Managed deployment of the company's software stack from the ground up, including release processes, production maintenance, and operational reliability.
- Improved search/Solr indexing performance by roughly 100x by building a custom Java indexing application.
- Built APIs and product features that helped customers collect, upload, process, transcribe, search, and manage video research data.
- Managed data migrations, backend maintenance, deployment automation, DevOps, production support, and release reliability in a two-developer engineering team.

Tango Group — Software Developer

Feb 2012-Oct 2012 | Boulder, CO

- **Context:** Digital agency/client-services environment.
- Implemented Ruby on Rails features, fixed bugs, wrote tests, and supported PostgreSQL-backed client product development.

TECHNICAL STACK

Languages: Rust, Python, Ruby, Go, TypeScript, SQL

Frontend: React, TypeScript, HTML, CSS, dashboards, browser extensions, admin tools

Backend: Ruby on Rails, FastAPI, APIs, WebSockets, background jobs, service architecture, CLI systems

Data: PostgreSQL, ClickHouse, SQLite, ETL, data modeling, event logs, analytics pipelines

Infrastructure: Kubernetes, Docker, AWS, Linux, Ansible, CI/CD, sysadmin, Grafana, monitoring

Domains: SaaS, internal tools, automation, analytics, research infrastructure, trading systems, data-heavy products

EDUCATION

University of Colorado Boulder — B.S. Computer Science, 2012